

14. Ectopic Pregnancy Treatment: Methotrexate, Surgery, and Expectant Management

Companion reading handout for simultaneous listening.

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The most dangerous ectopic treatment plan is the one that sounds complete on day 1 but has no reliable path to completion.

That is the central idea of this final lecture.

Ectopic treatment is not finished when the methotrexate injection is given. It is not always finished when the operation ends. It is finished when the patient is clinically safe and hCG has resolved.

So the dominant model is this: treatment is a longitudinal safety contract.

You choose the path by stability, rupture risk, contraindications, hCG behavior, fertility anatomy, patient preference, and follow-up reliability. Then you follow the path until the danger is actually gone.

Before any treatment choice, bring forward the safety rule from the previous lecture. Location before medication. Methotrexate should be used only for confirmed or high-suspicion ectopic pregnancy after intrauterine pregnancy has been reasonably excluded, unless the patient is unstable and needs surgery rather than a diagnostic debate.

Methotrexate

Methotrexate can be considered when the patient has a confirmed or high clinical suspicion of ectopic pregnancy, is hemodynamically stable, has an unruptured mass, has no absolute contraindication, and can participate reliably in follow-up.

That last phrase is not administrative. Follow-up reliability is part of the treatment. If the patient cannot return for hCG checks, cannot understand rupture precautions, or cannot access urgent care, methotrexate becomes less safe.

The mechanism explains both the power and the danger of the drug.

Methotrexate is a folate antagonist. It inhibits dihydrofolate reductase, which interrupts nucleotide synthesis, DNA synthesis and repair, and cell replication.

Trophoblast is rapidly dividing tissue, so methotrexate can stop the ectopic pregnancy from continuing to grow.

But trophoblast is not the only rapidly dividing tissue in the body. Bone marrow, gastrointestinal mucosa, respiratory epithelium, and other proliferating tissues can also be affected. That is why the contraindications are not random.

Do not give methotrexate when there is an intrauterine pregnancy. Do not give it to a patient with immunodeficiency, moderate or severe anemia, low white blood cells, low platelets, methotrexate sensitivity, active pulmonary disease, active peptic ulcer disease, clinically important liver dysfunction, clinically important renal dysfunction, breastfeeding, ruptured ectopic pregnancy, hemodynamic instability, or inability to participate in follow-up.

Notice the pattern. Bone marrow problems matter because methotrexate can suppress dividing marrow. Active lung or gastrointestinal disease matters because those tissues can be injured. Liver and kidney disease matter because methotrexate is hepatotoxic and renally cleared. Instability and rupture matter because a slow medical treatment cannot substitute for controlling hemorrhage.

Relative contraindications are different. They are warning lights, not always stop signs.

Relative contraindications include embryonic cardiac activity, high initial hCG, ectopic mass greater than four centimeters, and refusal to accept blood transfusion.

These features do not always make methotrexate impossible, but they predict lower success or higher consequence if failure occurs. High hCG is the cleanest example. In systematic review evidence, methotrexate failure was at least fourteen point three percent when pretreatment hCG was greater than five thousand, compared with three point seven percent when hCG was less than five thousand. Rapidly increasing hCG, especially more than fifty percent in forty eight hours, and fetal cardiac activity also predict failure.

So the practical sentence is this: methotrexate is most comfortable when the patient is stable, reliable, unruptured, low risk for failure, and truly has ectopic pregnancy rather than unresolved location.

The regimens

The single-dose regimen is the one residents most often need to carry clearly.

Day 1: methotrexate fifty milligrams per meter squared intramuscularly.

Check hCG on day 4 and day 7.

The hCG may rise by day 4. That can be normal. Do not call day 4 rise failure by itself.

The key response marker is the change from day 4 to day 7. If hCG decreases by more than fifteen percent from day 4 to day 7, continue weekly hCG until the level reaches the nonpregnant range for that laboratory.

If the decrease from day 4 to day 7 is less than fifteen percent, the risk of treatment failure is high. The next step is additional methotrexate or surgery, depending on symptoms, stability, diagnosis, and the clinical picture. If hCG does not decrease after two doses in the single-dose pathway, surgical management should be considered.

Here is the resident reflex: day 4 can rise; day 7 decides.

The two-dose regimen gives methotrexate fifty milligrams per meter squared on day 1 and again on day 4. hCG is checked on day 4 and day 7. If the day 4-to-seven decline is adequate, follow weekly. If it is not, another dose can be given on day 7 with reassessment on day eleven, and sometimes another dose on day eleven with reassessment on day fourteen. If hCG does not decline after four doses, surgery is considered.

The fixed multidose regimen is more intensive. Methotrexate one milligram per kilogram is given on days one, three, five, and seven, alternating with folinic acid zero point one milligrams per kilogram on days two, four, six, and eight. The folinic acid is rescue therapy to reduce methotrexate toxicity. Treatment continues until hCG falls by fifteen percent from the previous measurement, then weekly hCG continues until resolution.

There is no single universally perfect regimen. Observational studies put systemic methotrexate success roughly between seventy and ninety five percent, depending on patient selection, protocol, and starting hCG. The single-dose regimen is simplest and often appropriate for lower or plateauing hCG. The two-dose regimen may be considered when the initial hCG is higher and the clinician wants more efficacy than single dose while avoiding the adverse-effect burden of fixed multidose therapy.

The follow-up

After methotrexate, hCG must be monitored until it reaches a nonpregnant level. Resolution usually takes two to four weeks, but it can take up to eight weeks.

This is why the treatment is not the injection. The treatment is the injection plus the surveillance.

Ultrasound is not routinely used to prove resolution, because ultrasound findings do not reliably predict rupture or time to resolution. The mass may look persistent while trophoblastic activity is falling. hCG is the safety line.

Counseling after methotrexate must be concrete enough that the patient can act on it.

She must know rupture symptoms: worsening abdominal or pelvic pain, shoulder pain, dizziness, syncope, heavy bleeding, or feeling suddenly unwell. She needs to

know these symptoms require urgent care, not a routine phone call days later.

She should avoid folic acid supplements and folate-rich foods during treatment, because folate can oppose the mechanism of methotrexate. She should avoid nonsteroidal anti-inflammatory drugs, alcohol, vigorous activity, intercourse, unnecessary pelvic examinations, unnecessary ultrasound examinations, and excessive sunlight exposure until resolution. She should avoid narcotic pain medication when possible because it may mask worsening symptoms. She should also avoid gas-producing foods if they may confuse the interpretation of abdominal pain.

Some abdominal pain two to three days after methotrexate can occur from the cytotoxic effect on trophoblastic tissue. Vaginal spotting is also expected. But the teaching point is not to reassure every pain away. The task is to separate expected treatment effects from rupture.

Common adverse effects include nausea, vomiting, and stomatitis, especially with more dosing. Liver enzyme elevation can occur and usually resolves after stopping. Alopecia is rare at ectopic doses. Pneumonitis is rare, but fever or respiratory symptoms after methotrexate should be taken seriously.

Pregnancy after methotrexate also needs specific counseling. The patient should not become pregnant again until resolution is confirmed. FDA labeling recommends avoiding pregnancy during treatment and for at least one ovulatory cycle after therapy. Some experts recommend waiting at least three months after the last dose. Available limited evidence suggests methotrexate treatment for ectopic pregnancy does not adversely affect subsequent fertility or ovarian reserve, but counseling should be honest about the difference between limited reassuring evidence and unlimited certainty.

Surgery

Surgery is required when the patient has hemodynamic instability, symptoms of ongoing rupture, signs of intraperitoneal bleeding, an absolute contraindication to methotrexate, or failed medical management. Surgery should also be considered when relative contraindications make methotrexate less likely to succeed or when there is a concurrent surgical indication.

In stable patients, laparoscopy is typical. Laparotomy is usually reserved for unstable patients, large hemoperitoneum, or poor visualization.

For a stable nonruptured ectopic pregnancy, both intramuscular methotrexate and laparoscopic surgery can be safe and effective. The difference is not that one is always modern and the other is always aggressive. They solve different problems with different burdens.

Methotrexate avoids anesthesia and surgical risk, but it has lower success than laparoscopic salpingectomy in some comparisons and demands longer surveillance,

more visits, and more blood tests. Surgery gives immediate anatomic treatment, but has operative and anesthetic risk and requires a decision about the tube.

That brings us to salpingectomy and salpingostomy.

Salpingectomy removes part or all of the affected fallopian tube. It is generally preferred when the tube is severely damaged or bleeding. It can also be reasonable when future fertility is desired but the opposite tube is healthy.

Salpingostomy opens the tube and removes the ectopic pregnancy while leaving the tube in place. It should be considered when future fertility is desired and the opposite tube is damaged, because removing the affected tube may make spontaneous conception much less likely.

The tradeoff is persistent trophoblast. After salpingostomy, serial hCG is required to confirm resolution. If incomplete resection is a concern, a single prophylactic dose of methotrexate may be considered.

The fertility data are nuanced. Randomized trials comparing salpingectomy and salpingostomy did not show a statistically significant difference in subsequent intrauterine pregnancy or repeat ectopic pregnancy. Cohort data suggest salpingostomy may have a higher subsequent intrauterine pregnancy rate, but also a higher repeat ectopic risk, ten percent versus four percent, compared with salpingectomy.

The clinical translation is not to memorize one universal procedure. The clinical translation is to inspect the tube, inspect the opposite tube, ask about future fertility, and match the operation to anatomy and goals.

Expectant Management

Expectant management is not doing nothing. It is active observation of a pregnancy that is already showing objective evidence of resolution.

Candidates should be asymptomatic, clinically stable, reliable for follow-up, and willing to accept the risks, including tubal rupture, hemorrhage, and emergency surgery. There should be objective evidence of resolution, usually plateauing or declining hCG.

The lower the hCG, the more plausible spontaneous resolution becomes. When the initial hCG is less than two hundred, about eighty eight percent of patients experience spontaneous resolution. Higher levels have lower spontaneous resolution rates.

Abandon expectant management for intractable pain, significantly increasing pain, insufficient hCG decline, or signs of rupture. In other words, expectant management is safe only while the patient's biology and symptoms continue to justify it.

Bring the whole course together

Across office gynecology and early pregnancy care, the same discipline keeps repeating.

Do not treat symptoms as diagnoses. Classify abnormal bleeding before choosing therapy. Diagnose vaginitis by ecosystem and organism, not discharge appearance. Treat prolapse by symptoms and function, not anatomy alone. Read adnexal ultrasound as architecture of risk. Diagnose early pregnancy loss with conservative thresholds. Locate pregnancy before methotrexate. And after ectopic treatment, follow until hCG resolves.

The safest clinician is not the one who knows the most isolated lists. It is the one who knows which distinction changes management and refuses to skip that distinction when the clinic is busy, the patient is anxious, or the team wants a shortcut.

Final Resident Reflex

Surgery treats anatomy. Methotrexate treats trophoblast. Expectant management follows biology that is already resolving.

Choose the path by stability, rupture risk, contraindications, hCG behavior, fertility anatomy, patient preference, and follow-up reliability.

Then finish the treatment.

The danger is not over when the first intervention is done. It is over when the patient is safe and hCG has resolved.